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The role of renewable energy consumption and institutional quality level in promoting environmental sustainability among developing countries: A dual pathway to unveil environmental Kuznets curve (EKC) dynamics

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A R T I C L E I N F O

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A B S T R A C T

Economic growth in developing economies often coincides with rising environmental degradation, leading to heated debate over whether current growth pathways are environmentally sustainable. While previous studies have examined the nexus between growth and emissions, limited attention has been paid to determining the threshold level beyond which growth improves environmental quality, or to explaining how renewable energy consumption and institutional quality shape this pathway. This study, therefore, addresses this gap by examining the GDP per capita threshold at which CO_2 emissions begin to decline, while assessing whether renewable energy consumption and institutional quality mediate and moderate this relationship. Using panel data from 77 developing countries spanning 2000-2023, the study employs the cross-sectional augmented autoregressive distributed lag (CS-ARDL) and dynamic panel threshold (kink) models as the main econometric approaches. The findings indicate that renewable energy consumption plays a mediating role in explaining the growth-emission nexus among developing countries. Second, there exists a threshold level of GDP per capita ($4230.18), beyond which its further growth significantly reduces CO_2 emissions. Third, that institutional quality moderates the nexus between growth and emissions by accelerating the attainment of the EKC turning point, estimated at $2.465.13. The findings highlight the importance of scaling up investment in renewable energy and strengthening institutional quality frameworks (political, economic, and regulatory institutions) to align economic growth with environmental sustainability. Policies geared towards the development of human capital, such as education and skills development, are essential to help developing countries surpass the identified threshold level and improve environmental quality.

1. Introduction

The IMF report on environmental quality has projected that the unchecked effects of climate change could reduce global GDP by 11-15%, with developing countries standing to lose up to 19% of their output by 2030 (Hicklin). The report comes amid the struggle of developing countries to balance environmental sustainability and economic growth, following their significant carbon emissions into the global space and their high vulnerability to climate change shocks (Malik et al., 2025, pp. 1–31). Sub-Saharan African countries, for

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instance, have seen a nearly 200% increase in carbon dioxide emissions, rising from 100 kilotons in 2000 to 300 kilotons in 2020 (Science, 2022). Rapid urbanization and industrialization, in the context of weak institutions, have historically prioritized short-term economic gains over ecological protection, thereby linking growth to environmental degradation (Georgescu, 2025). Within this context, access to renewable energy and institutional reinforcement have emerged as key drivers of sustainable economic growth, especially where ecological imperatives need to be reconciled with development (Dinda, 2004). This dual mechanism provides a strategic advantage for aligning environmental and economic growth objectives, especially in the Global South, which is projected to continue to experience higher impacts of environmental degradation.

The long-standing EKC hypothesis posits that environmental degradation often corrects itself once an economy surpasses a certain critical income level, so countries should strive to achieve this level (Dinda, 2004). The hypothesis posits that the relationship between economic growth and environmental quality is non-linear, distorting in the initial stages of growth but improving after a certain income threshold is reached. This yields an inverted U-shape relationship between these variables, economic growth and environmental quality (Beşe & Kalayci, 2019). While the existing evidence supports this pattern among developed countries, its validity in developing countries remains unsettled, with the EKC turning point often delayed or even absent (Masron & Subramaniam, 2020). Recent studies suggest that income growth alone cannot sufficiently explain environmental outcomes, as other structural factors play a significant role in shaping the EKC path (Georgescu, 2025; Masron & Subramaniam, 2020).

Among prominent structural factors, renewable energy consumption and level of institutional quality have emerged as key channels through which environmental degradation can be decoupled from economic growth (Shayanmehr et al., 2023). Recent studies in this vein have found that an increase in renewable energy consumption reduces carbon emissions and weakens the positive nexus between economic growth and pollution (Mahmood et al., 2023). Similarly, the institutional quality level, which captures the rule of law, governance effectiveness, policy credibility, and regulatory capacity related to environmental conservation, is a key factor influencing growth and environmental outcomes. For instance, Sahin et al. (2025) found that strong institutional quality enhances the enforcement of environmental laws that promote accountability and innovation, leading to reduced emissions. Empirical studies such as Ikechukwu and Agu (2024) and Oje (2024, pp. 42–64) have shown that countries with higher levels of institutional quality tend to experience transmission towards sustainable growth at lower emission levels than their low-institutional counterparts. Importantly, the level of institutional quality can influence the EKC by accelerating the income level at which environmental improvement begins along the development process, which can be explained by the moderating effect (Sahin et al., 2025). On the other hand, renewable energy infrastructure, such as hydro, wind, and solar, presents greater opportunities for developing countries on their journey to decarbonize the energy system. For instance, Mustafa et al. (2022) suggest that increased renewable energy consumption stimulates long-term energy security and sustained growth.

Therefore, this study sought to investigate the role of institutional quality and renewable energy consumption in shaping the nexus between growth and environmental quality among developing countries. Through the construction of a scientific theoretical framework and the estimation of a dynamic threshold empirical model, this study seeks to make the following three distinct contributions: First, it determines the income (GDP per capita) threshold that developing countries need to start experiencing environmental improvement under the EKC hypothesis. Second, this study examines whether renewable energy consumption explains the mechanism through which growth translates to environmental improvement among developing countries (mediating role). Third, it examines whether institutional quality level helps accelerate the achievement of the income threshold required for developing countries to start experiencing environmental improvement (EKC turning point).

Unlike previous studies that have examined the EKC hypothesis in isolation (see Adu & Denkyirah, 2017; Beşe & Kalayci, 2019; Georgescu, 2025; Mahmood et al., 2023), this study explicitly integrates renewable energy consumption and institutional quality into a growth-environmental quality analysis across selected countries. It posits that stronger institutional quality accelerates the attainment of the ECK turning point. At the same time, renewable energy consumption propels economic growth towards low-emission levels by reducing dependence on fossil fuels. By considering these variables, this study provides empirical evidence of their role in shaping the direction and speed of growth and environmental quality pathways.

2. Literature review

2.1. Theoretical literature review

This study is anchored on three main theories as discussed.

2.1.1. Environmental Kuznets Curve (EKC) theory

The EKC theory, first proposed by Grossman and Krueger (1991), posits an inverted U-shaped relationship between economic development and environmental quality, whereby emissions tend to increase in the early stages of growth and then decline once a certain income threshold is reached. According to this theory, the inverted U-shaped relationship between economic growth and environmental quality arises because, in the early stages of growth, countries tend to rely on poor and inefficient production technologies that exacerbate environmental damage through greenhouse gas emissions and pollution. However, at high levels of economic growth, the demand for environmental quality, combined with technological improvements, increases significantly, reducing emissions and pollution (Aghion & Bolton, 1997). Similarly, environmental degradation intensifies during the early stages of economic growth due to overdependence on primary and secondary production sectors, such as agriculture and manufacturing, which are environmentally pollution-intensive. Limited awareness, weak institutional capacity to develop prerequisite infrastructure while enforcing environmental policies, and mounting urbanization pressures are widely recognized as recent key drivers of environmental degradation in the early stages of economic growth (Huang et al., 2024). Although relevant, the EKC hypothesis has been criticized for oversimplifying complex, country-specific dynamics and assuming that all countries have similar growth trajectories (Sarkodie & Strezov, 2019). This may not be the case for developing countries with varying institutional levels, governance, and access to clean technology. This shortcoming prompted the current study to examine whether the relationship between economic growth and environmental quality is influenced by institutional quality and renewable energy consumption.

2.1.2. Sustainable development theory

This is an environmental conservation theory that emerged in the 1980s, following the discussion of the World Commission on Environment and Development (WCED) report of 1987, commonly referred to as the Brundtland Report (Imperatives, 1987). The theory defines sustainable development as one that meets the present economic, social, and environmental needs without compromising the ability of future generations to meet their own needs. This presents a paradigm shift from traditional growth models analysis, which rarely emphasized striking a balance among economic growth, social equity, and environmental sustainability. The main premise of sustainable development theory is that development is important if it occurs within the planet's ecological limits, recognizing the existence of present and future environmental needs (Ekins, 1993). The sustainable development theory also emphasizes the consideration of intra- and intergenerational equality in economic development and climate change analysis, advocating for fair distribution of opportunities and resources both across and within generations (Naciti, 2019). Over time, the sustainable development theory has evolved to include more specific economic framework analysis, such as the United Nations Sustainable Development Goals (SDGs) agenda 2030, extending its application across scales and sectors of interest (Josephsen, 2017). In line with the current study, this theory provides a crucial lens for analyzing the nexus between economic growth and environmental degradation, by determining whether this relationship respects environmental degradation thresholds as the economy expands. In particular, the sustainable development theory supplements the EKC hypothesis by considering not only whether environmental degradation decreases with income growth but also the direction, quality, and sustainability of economic growth (Ekins, 1993).

2.1.3. Ecological modernization theory

The Ecological Modernization Theory (EMT) offers a valuable theoretical framework for examining the nexus between economic growth and environmental quality (Buttel, 2000). The theory is particularly relevant in the analysis of developing countries due to structural transformation and high vulnerability to environmental degradation. EMT was first put forward by Joseph Huber and Martin Jänicke in the 1980s as a response to the need to address social and environmental concerns in Western European countries, particularly Germany, the United Kingdom, and the Netherlands (Huber & Huber, 2010). Since then, the theory has continued to develop, with critics of the traditional ecological pessimism theory proposing that environmental sustainability and economic growth are not inherently contradictory but can move together following technological innovation and proactive environmental policies (Palestine, 2015). The central proposition of EMT is that there exists a certain level of economic and institutional development beyond which society shifts from pollution- and resource-intensive modes of production towards cleaner, more efficient, and less carbon-intensive production systems (Buttel, 2000). The achievement of this critical point of economic development is made possible by better environmental regulations, the advancement of green technologies, and the green market enabled by the provision of eco-friendly goods and services (Mustafa et al., 2022). The main proposition of ecological modernization theory is thus that, after crossing certain development thresholds, nations acquire the technical capacity to decouple economic growth from environmental degradation. This allows continued development without increasing ecological damage (Palestine, 2015). In line with the current study, the EMT provides a theoretical framework for investigating the existence of a non-linear threshold relationship between economic growth and environmental degradation indicators, particularly carbon emissions. Through a theoretical analysis, the EMT provides a basis for identifying the mechanisms by which the threshold point in the nexus between economic growth and environmental degradation is studied, as well as for selecting other models that control for individual countries' characteristics.

2.2. Empirical literature review

Empirically, the relationship among institutional quality, renewable energy consumption, and environmental sustainability has continued to be examined within the EKC framework at varying levels of scope. Varying results have been found depending on the country of analysis, the scope, and the methodology employed. For instance, previous studies such as Ozkan et al. (2023), Vo et al. (2024), Asiedu and Aboagye (2025), Gil-p et al. (2023), and Ngozi et al. (2023) have shown that economic growth can intensify pollution, leading to a late or even no realization of the EKC turning point at all. This contradicts the invested U-shaped traditional EKC proposition, implying that a rise in income level does not automatically translate into environmental improvement in some economies. Similarly, some studies have revealed that environmental response to economic growth is asymmetric and non-linear. This means that technological and economic shocks tend to have varying effects on emissions, rendering the linear examination of the nexus between economic growth and environmental quality insufficient (Ullah et al., 2024). Some studies, such as Azimi and Rahman (2024) and Khan et al. (2022), have also shown that methodological choice can strongly influence findings on environmental sustainability due to the uneven impacts of economic and institutional shocks, which often remain hidden in linear model analyses. This implies that employing advanced nonlinear models can accurately capture the environmental transitions and EKC dynamics.

Some existing studies have shown that economic growth indirectly influences environmental quality through renewable energy consumption (Khalid et al., 2021). In particular, previous studies such as Aslan et al. (2021), Zafar et al. (2021), and Hou et al. (2023) suggest that renewable energy consumption contributes to the improvement of environmental quality by lowering carbon emission and dependence on fossil fuel consumption, leading to the acceleration of the EKC turning point, which is premised on the growth and environmental quality nexus. Furthermore, studies in developing countries have shown that renewable energy consumption, coupled with institutional quality, reduces carbon emissions (Khalid et al., 2021). This implies that effective governance enhances policy enforcement and regulatory implementation that promote investment in technologies, promote clean energy consumption, and improve environmental quality, even as economic growth is realized (Islam, 2021; Usman et al., 2022). Further, previous studies reinforce the complementary relationship between environmental sustainability and institutional quality. For instance, Wang et al.'s (2022) study on the BRICS and other emerging countries found that green energy transition, innovation, and governance quality collectively influence environmental performance. In particular, this study found that institutional effectiveness promotes environmental policy compliance and continuity, thereby strengthening the role of economic growth, either directly or indirectly, in stimulating a long-run reduction in emissions. These findings show that environmental sustainability in developing countries does not depend on a single factor but on several interrelated variables that jointly shape outcomes. Therefore, the traditional treatment of institutional quality and renewable energy consumption as control variables in environmental analysis is insufficient, and further studies should consider a dual-pathway relationship between these variables, including moderating and mediating mechanisms. These shortcomings motivated the current study to analyze the nexus among renewable energy consumption, economic growth, institutional quality, and environmental sustainability as interconnected variables through the EKC framework.

The reviewed literature presents mixed evidence, with some studies supporting an inverted U-shaped relationship, while others propose a direct monotonic deterioration of the environment as the economy grows. Recent studies have further revealed that, beyond the growth effect, a country's level of institutional development plays a critical role in shaping environmental quality outcomes through the proper design and enforcement of environmental policies. In parallel, the increase in renewable energy consumption has also emerged as a critical pathway for decoupling economic growth from pollution and promoting sustainable growth. Building on these insights, this study advances the existing literature by re-examining the EKC pathway in developing countries, using institutional quality and renewable energy consumption as moderating and mediating factors in the growth-environmental quality nexus, rather than the traditional analysis, which has largely treated these factors as control variables. Furthermore, this study employs an advanced non-linear dynamic model that improves upon previous studies that estimated the relationship between these variables using the averaging technique. This helps identify the critical level of institutional quality and renewable energy consumption levels needed to stimulate improvements in environmental quality. The dynamic model employed in this study further accounts for the evolutionary nature of environmental conditions over time, a common phenomenon in both developing and developed countries (Zhao et al., 2025).

3. Methodology

3.1. Study scope and data

This study uses a balanced global panel dataset covering 77 developing countries over 24 years, from 2000 to 2023. The data was collected from the World Development Indicators, hosted by the World Bank. 77 out of the possible 154 countries were sampled based on data availability. A detailed summary description of study variables, Tables 1 and 2.

3.2. Variable selection

The dependent variable in this study was CO2 emissions. This acted as a proxy for environmental degradation. GDP per capita was the main independent variable used to explain the EKC. Renewable energy consumption was added to capture the energy transition hypothesis. Trade openness and FDI inflows represented global factors influencing environmental outcomes, as posited by the pollution halo and pollution haven hypotheses (Mahmood et al., 2023; Ozkan et al., 2023). Institutional quality was also included in line with institutional theory, which holds that effective governance determines the success of environmental policy. The selection of these study variables is reinforced by prior research (see Khan et al., 2022; Lee et al., 2020, pp. 41776–41786; Zhang et al., 2023) that has used similar variables in their analyses. For instance, Dina et al. (2021) argue that foreign direct investment and trade openness influence environmental quality by transferring pollution and clean technologies. Institutional quality and renewable energy

Table 1
Variables description and data source.

Symbol	Definition	Measurement unit	Data source	Expected sing
CO2	Carbon dioxide emission	Metric tons per capita	Global Footprint Network	
GDPP	Gross domestic product per capita	GDP per capita (constant 2015 US$)	World Development Indicators (WDI)	+/− (EKC hypothesis)
FDI	Foreign direct investment inflow	Net FDI inflow as % of GDP	World Development Indicators (WDI)	+
RE	Non-renewable energy consumption	% of total energy consumption	World Development Indicators (WDI)	-
INST	Institutional quality	Average composite index of six institutional dimensions (*Voice and accountability, political Stability, government effectiveness, regulatory quality, rule of law, and corruption control*)	World Indicators Governance Indicators (WGI)	-
TR	Trade openness	Sum of export and import as a % of GDP	World Development Indicators (WDI)	+

Table 2
Descriptive statistics.

Statistic	lCO2	GDPP	FDI	INST	REG	TRA
Mean	16.64	7.55	3.43	34.29	43.97	69.83
Maximum	23.20	10.11	55.07	77.68	98.30	220.41
Minimum	11.89	4.69	−37.17	1.52	0.00	15.68
Std.dev	2.08	1.15	4.93	17.97	32.06	17.97
Skewness	0.34	−0.05	3.56	0.27	0.16	0.19
Kurtosis	2.94	2.09	32.69	2.42	1.51	4.53

consumption have been reported to shape the relationship between growth and emissions. These can serve as moderating or mediating variables rather than just control variables (Huang et al., 2024). By integrating these variables, this study aligns with the prior literature and provides a comprehensive framework for explaining the growth-emission path.

3.3. Analytical framework and research methodology

3.3.1. Analytical framework

Empirical investigation in scientific research focuses on selecting appropriate variables from the available dataset, guided by relevant theories and prior empirical studies in the area of interest. Therefore, this study formulated an analytical model informed by prior studies such as, (Christoph Martin and Lieb, 2003; Dinda, 2004; Grossman & Krueger, 1991). In particular, the current study extended on the empirical models by Sahin et al. (2025) and Shayanmehr et al. (2023) to formulate a theoretical model on determinants of environmental quality, as follows:

$$CO_2 = (GDPP, X') \tag{1}$$

CO2 emissions serve as a proxy for environmental quality, while GDP per capita denotes gross domestic product per capita, the main independent variable that captures the role of economic growth. X is a set of control variables, as well as other growth-emission transmission variables, such as foreign direct investment inflow, trade openness, institutional quality, and renewable energy consumption (Khan et al., 2022; Lee et al., 2020, pp. 41776–41786). An extension of Equation (1) is therefore given as;

$$ln\ CO_{2it} = \beta_1\ ln\ CO_{2it-1} + \beta_2 lnGDPP_{it} + \beta_3 lnGDPP_{it}^2 + \beta_4 FDI_{it} + \beta_5 TR_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{2}$$

Where l $ln\ CO_{2it}$, and $ln\ CO_{2it-1}$ are natural logs of current and lagged carbon dioxide emission, $lnGDPP_{it}$ represents gross domestic product per capita. At the same time, $lnGDPP_{it}^2$ is the squared values that measure the nonlinear relationship between variables. FDI_{it} and TR_{it} denote the control variables foreign direct investment inflow and trade openness, respectively. The country fixed effect and time effect are donated by μ_i and $\lambda_{t,}$ respectively.

3.4. Research methodology

To study the relationship between environmental sustainability and economic growth, this study employed the EKC model and conducted two alternative econometric methodologies. In the first estimation, the cross-sectional augmented distributed lag (CS-ARDL) technique was used to test the short- and long-run relationships between economic growth and environmental sustainability, and to examine the mediating effect of renewable energy consumption in this relationship. This was followed by the dynamic panel threshold estimation model to test whether there is a threshold level of economic growth beyond which environmental quality improves.

3.4.1. Cross-sectional augmented autoregressive distributive lag estimation

The CS-ARDL model was employed as the baseline model in this study. The model was selected because of its ability to accommodate the omitted variable biases while simultaneously determining the growth of regressors which are integrated of either order I(0) or I(1). Furthermore, the cross-sectionally augmented distributed lag is always reliable. It remains valid in the presence of strong cross-sectional dependence and common unobserved global shocks, both of which are ignored by traditional panel estimators. However, they can produce biased and inconsistent results (Fofana N'Zue, 2020). The CS-ARDL technique was employed in this study because it efficiently estimates short- and long-run relationships in panel data while also addressing the problem of cross-sectional dependence, which is common among macroeconomic variables (Hou et al., 2023). The techniques allow heterogeneous coefficients across panels in the short run while imposing homogeneous equilibrium in the long run, yielding efficient estimates. Although the current study used large panels (N > T), the CS-ARDL still remains appropriate, because the time series T = 24 is moderately large to capture both short and long-run coefficient as provided by (Ditzen, 2018).

Equation (2) was therefore estimated using the CS-ARDL model through the pooled mean group (PMG) technique. This technique accommodates both short-run and intercept coefficients and yields consistent results when coefficients are averaged across countries. Theoretically, the EKC hypothesis provides for a long-run relationship between economic growth and environmental quality. In the context of the current study, while developing countries may tend to have diverse short-run structures in terms of regulatory capacity, institutional frameworks and income level, the underlying long-run relationship between economic growth and emissions is expected to converge in the long run (Sahin et al., 2025).

The mean group vector is given by $\sigma = E(\sigma_i)$, where the individual level or long-run coefficients are assumed to be equal to $\sigma = \sigma_i$, and are given as;

$$\sigma i = \frac{\sigma_{0i} + \sigma_{1i}}{1 - \beta_1}$$

Theoretically, the endogenous growth and EKC hypotheses hold that income growth tends to influence environmental quality indirectly by altering technological adoption, energy consumption, and energy efficiency. This means that as the economy expands, it enables investment in clean energy, making renewable energy consumption an intermediate pathway that links economic growth to environmental quality rather than a control variable in analysis (Hu & Wang, 2025). Similarly, previous empirical studies have shown that economic growth often influences outcomes through intermediate mechanisms, such as human and institutional processes, which confirms the importance of transmission analysis when dealing with growth variables (Botta, 2020). Therefore, building on Equation (2), the test for mediating effect is tested methodologically using the following procedure: (i) Regressing the main independent variable (GDPP) on the mediating variable (RE) as in Equation (3) to check whether their relationship is significant. (ii) Regression of renewable energy on environmental quality alongside GDP per capita and other control variables as shown in Equation (4). The mediation effect of renewable energy is present when GDP per capita growth significantly influences renewable energy consumption, while environmental quality is also influenced by renewable energy consumption (Ojeka et al., 2024).

$$RE_{it} = \varphi_0 + \varphi_1 lnGDPP_{it} + \varphi_2 lnGDPP_{it}^2 + \mu_i + \lambda_t + \varepsilon_{it} \tag{3}$$

$$ln\ CO_{2it} = \pi_0 + \pi_1\ ln\ CO_{2it-1} + \pi_2 lnGDPP_{it} + \pi_3 lnGDPP_{it}^2 + \pi_4 FDI_{it} + \pi_5 TR_{it} + \delta RE_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{4}$$

Where, φ_0 and π_0 are the constant terms while $\varphi_1, \varphi_2\ \pi_1, \pi_2, \pi_3 \pi_4$ and δ denote the regression coefficients. This tests the three steps proposed by Baron and Kenny (1986) for validation. First, in the baseline model (Equation (2)), it is expected that both β_2 and $\beta 3$ be significant, irrespective of their sign. Second, from Equation (3), which estimates the relationship between economic growth and renewable energy consumption, $\phi 1$ and $\phi 2$ are expected to be equally significant. Finally, to demonstrate the mediating effect of renewable energy consumption on CO_2 emissions, it is expected that $\pi 2$, $\pi 3$, and δ are all statistically significant, and that $\beta 2$ and $\beta 3$ are greater than $\pi 2$ and $\pi 3$, respectively.

3.4.2. *Dynamic panel threshold estimation*

Given that economic growth is inherently dynamic, it often alters environmental outcomes over time. The Environmental Kuznets Curve theory posits that the nexus between environmental quality and economic growth is not constant but varies across income levels. Therefore, estimating static panel models may not adequately capture the dynamic relationship between these variables (Q. Hu & Wang, 2024). Therefore, to capture the potential threshold relationship between environmental quality and economic growth, a dynamic panel threshold regression model was employed to test for nonlinearity while accounting for the dynamics of both variables. This technique was chosen for this analysis because it can identify a threshold in economic growth beyond which environmental quality improves (Alemu et al., 2023). Therefore, following Topal (2014), this study modified the baseline equation in 2 to a dynamic panel threshold, as in Equation (5).

$$lnCO_{2it} = \pi_0 + \eta_1 X_{it}'.1(q_{it} \leq \gamma) + \eta_2 X_{it}'.1(q_{it} > \gamma) + \mu_i + \varepsilon_{it} \tag{5}$$

Where lnCO2 denotes the natural logarithm of current carbon dioxide emission, $X'it$ is a set of control variables including the lagged dependent variable, q_{it} is the threshold variable (GDP per capita), γ is the endogenously determined threshold value, and 1(.) is the indicator function.

The inclusion of $lnCO2it - 1$ introduces endogeneity in the panel estimation model. This makes the dynamic panel threshold model by Kremer and Nautz (2013) inefficient when slopes are heterogeneous. Therefore, this study adopted the extended model by Seo and Shin (2016), which uses the dynamic panel threshold (kink) technique, allowing the inclusion of lagged dependent variables and endogenous covariates in the model. This model is based on first-difference GMM estimators that include linearity tests for threshold effects (Seo et al., 2019). The proposed threshold model typically exhibits a discontinuity in the estimated model; however, Seo et al. (2019) argue that it may be a kink rather than a discontinuity or jump.

Following the assumption by Seo et al. (2019), Equation (5) is further modified to accommodate the endogenous covariate and the kink restriction as follows.

$$lnCO_{2it} = \pi_0 + (1, X_{it}')\eta_1 1(q_{it} > \gamma) + \eta X_{it}' + \mu_i + \varepsilon_{it} \tag{6}$$

Where the individual effects μ_i and the incidental parameter η_1 are removed through first difference GMM transformation to give the final threshold estimation model as in Equation (7).

$$lnCO_{2it} = \pi_0 + k(q_{it} - \gamma)1(q_{it} > \gamma) + + \eta X_{it}' + \varepsilon_{it} \tag{7}$$

where k(.) denotes the kinked restrictions in the model, i = 1,2 … n and t = 1,2 … T. T is assumed to be fixed while n is as large as possible (n > T). Equation (7) forms the dynamic panel threshold kink model which estimated to uncover the threshold relationship between economic growth and environmental quality.

To test whether institutional quality has potential of accelerating attainment of Environmental Kuznets Curve turning point (moderating effect). Methodically, moderating effect of institutional quality on nexus between growth and environmental quality was test using the interaction model estimation. In this case, a panel regression analysis was conducted in which environmental quality was regressed on GDP growth per capita, institutional quality, and their interaction term (GDP per capita*institutional quality), alongside the study control variables. The coefficient of the GDP per capita*institutional quality variable is expected to be significant to confirm the moderating effect (Ogbuabor et al., 2023).

Therefore, Equation (7) is further transformed to 8 as;

$$lnCO_{2it} = \pi_0 + k(m_{it} - \rho)1(m_{it} > \rho) + + \eta X'_{it} + \varepsilon_{it} \tag{8}$$

Where m_{it}, is the new threshold variable (GDP per capita*institutional quality), while ρ is the threshold value. The remaining variables remain as described in Equation (5). From this equation, the moderating effect holds if the estimated threshold value in Equation (7) (γ) is greater than that ρ estimated in Equation (8).

To test the threshold significance, the Hansen's (1999) bootstrap test was employed. Test bootstrap methodology serves a dual purpose as test for threshold effect in the data and test whether the threshold effect is significant that is effect below threshold value is different from above. The test provides that, at the 95% confidence level, the confidence intervals should not overlap for the threshold values to be statistically different (Hansen, 1999).

4. Results and discussion

4.1. Descriptive statistics of the data

Table 2 presents descriptive statistics for the data. The descriptive statistics show moderate variation in the study variables, except for foreign direct investment inflow, which has a high standard deviation, implying very high volatility in its inflow into the selected countries. This can be attributed to their high sensitivity to the global economic circles, such as interest rate volatility and weak institutional quality (Deniz et al., 2021). Similarly, the small values of Skewness and Kurtosis imply a normal distribution across variables, except for FDI inflow, which shows a symmetric distribution (Hatem et al., 2022).

4.2. Correlation analysis

The non-normality of some variables in this study, as presented in Tables 2 and is expected in panel data analysis due to unobserved heterogeneity and specific country effects (Ertugrul & Demir, 2018). Table 3 presents Pearson pairwise correlation test results. Notably, three patterns emerge in the selected sample: GDP per capita is strongly positively correlated with CO2 emissions (r = 0.60), suggesting that high-income levels in developing countries are associated with high emissions (Lee et al., 2020, pp. 41776–41786). Second, renewable energy consumption (RE) has a strong negative correlation with CO2 emissions (−0.64) and a strong positive correlation with GDP per capita (0.60), which implies a crucial role for this variable in the growth-emission nexus among developing countries (Pesaran, 2021). Third, institutional quality has a strong positive correlation with GDP growth (0.57) and weak negative correlated with the emission (CO2) (−0.25). This suggests that institutional quality reduces emission through economic growth (Curto & Pinto, 2011). Notably, the test for multicollinearity in Table 3 provides variance inflation factor (VIF) values below 2, further justifying that there is no serious multicollinearity problem that may affect the test power of the independent variable coefficients (Curto & Pinto, 2011). These variations in results motivated this study to analyze the non-linear relationship between growth and emission through EKC analysis, test for the mediating role of renewable energy consumption, and determine whether institutional

Table 3
Correlation.

	CO2	GDPP	FDI	INST	REG	TRA
CO2	1.00					
GDPP	0.60	1.00				
FDI	−0.07	0.03	1.00			
INST	−0.25	0.57	0.05	1.00		
REG	−0.64	0.61	−0.08	−0.40	1.00	
TRA	0.03	0.26	0.26	0.30	−0.31	1.00

Test for Multicollinearity

	VIF	1/VIF
CO2	1.69	0.592
GDPP	1.59	0.629
FDI	1.07	0.934
INST	1.48	0.676
REG	1.54	0.649
TRA	1.11	0.900
Mean	1.413	

quality moderates the relationship between growth and CO_2 emissions.

4.3. Panel test for cross-sectional dependency

The panel test for cross-sectional dependency (CD) is another important test in panel data analysis. While it is generally assumed that disturbances in the panel data model are cross-sectionally independent for short panels ($N > T$), the reality is that cross-sectional dependency cannot be ruled out in panel data analysis (Pesaran, 2021). This study estimated the Pesaran CD test for cross-sectional dependency and presented the results in Table 4. From the test, it can be noted that cross-sectional pendency exists among panels as evidenced by the rejection of the null hypothesis of cross-sectional independence at 5% significance level. The cross-sectional dependency results imply that the study variables are highly heterogeneous; thus, specific country considerations when analyzing the response shock in this study might lead to biased conclusions regarding developing countries.

4.4. Panel unit root test

Due to cross-sectional dependence among panels, the second-generation unit root test was conducted, which accounts for CD (Khalid et al., 2021). Cross-sectionally augmented Im, Pesaran, and Shin (CIPS) and cross-sectionally augmented Dickey-Fuller (CADF), proposed by Pesaran (2007), were therefore tested both at the level and in first differences. The results are presented in Table 5. The result shows that some variables had unit roots at the level but all became stationary at the first difference, which accounts for cross-sectional dependency. This prompted the test for cointegration using the Westerlund (error-correction) panel cointegration technique before estimating the relationship between the study variables using the cross-sectionally augmented autoregressive distributed lag (CS-ARDL) model. The test reveals a long-run relationship between study variables which validates the estimation of CS-ARDL model proposed by (Chudik, Mohaddes, Pesaran, & Raissi, 2015). The model was selected because it is flexible to mixed conitegration I(0) and I(1) as implied by the second generation unit root test.

4.5. CS-ARDL model results

The cross-sectionally augmented autoregressive distributed lag (CS-ARDL) model was employed to estimate Equation (2) to uncover the short- and long-run relationships between CO_2 emissions and other variables. Equation (2) is estimated by including control variables such as institutional quality, foreign direct investment, and trade openness, and the results are presented in Table 6. From the results, the coefficient on the error correction term (ECT) is negative and significant, implying that any short-run shocks in the system will always revert to the long-run equilibrium (Bashir Jama, 2021). Importantly, the long-run relationship between GDP growth per capita and emissions is positive and significant, while the squared coefficient of per capita growth is negative. This means that initial growth in developing countries accelerates emissions, while higher levels of GDP per capita turn growth-emission downward. A plausible explanation for this result is that most developing economies' growth is mainly powered by urbanization and industrialization, which intensively consume fossil fuels; stringent environmental policies; and the transition to clean energy, which together fuel emissions in the early stages of growth. However, this is corrected as income levels increase, which is also consistent with the EKC hypothesis, which posits that growth in its initial stages stimulates environmental degradation until a certain high-income level, after which there is a shift toward greener energy development that tends to lower pollution (Beşe & Kalayci, 2019). Similarly, this study found a positive relationship between one-year lagged and current CO_2 emissions. This is expected because energy infrastructure, industrial and transportation processes, and systems do not change frequently. It is therefore expected that high pollution in previous years is followed by high pollution in the subsequent year (Adu & Denkyirah, 2017). The second lag of emission, however, has a negative, significant coefficient, which implies that while emission builds on its past level, it is partially corrected in the long run. This is possibly due to environmental policy adjustment, increased natural absorption, and technological advancement that corrects emissions in the long run (Nguyen & Duong, 2025).

Regarding the control variables, Table 6 further shows that trade openness and FDI inflow have positive, significant effects on emissions in developing countries. This could be because trade openness often increases industrial and production activities to meet growing export demand in developing countries, leading to increased energy consumption and pollution (Huang et al., 2024). Trade openness also reinforces the Pollution Haven Hypothesis, in which pollution-intensive firms and industries tend to relocate to developing countries with weak environmental regulatory frameworks, further increasing CO_2 emissions and pollution (Ozkan et al., 2023). The positive effect of trade openness on environmental pollution is in line with previous empirical studies such as Nguyen (2022), Khalid et al. (2021), Bakri et al. (2025), and Mai et al. (2024), who also found similar results. To add on, the positive relationship between FDI inflow and CO_2 emission could be due to the fact that an increase in foreign direct investment inflow into the

Table 4
Pesaran CD test for cross-sectional dependency.

Panel test for cross-sectional dependency	7.526***
Average absolute values of off diagonal elements	0.426
Probability values	0.000
F(76, 1766)	229.74
Prob > F	0.000

Note. *** denote level of significance at all confidence levels.

Table 5
Second-generation panel unit root test.

	Intercept		Intercept and trend	
	Level	First dif.	Level	First dif.
CADF test				
CO2	−2.521	−3.322***	−2.246	−2.912*
GDPP	−2.077	−3.101*	2.261	−3.247**
FDI	−1.719	−3.906***	−1.788*	−3.244*
INST	−1.089	−3.361**	−1.691	3.335**
REG	−1.441	−2.558***	−2.216	−4.818*
TRA	−2.303	−4.189**	−2.222	−3.434*H
CIPS test				
CO2	−2.032***	−5.118**	−2.626	−5.546**
GDPP	−2.306**	−4.611*	−2.249	−4.408*
FDI	−1.891	−3.928**	−1.974	−4.158*
INST	−2.233**	−5.281*	−2.686*	−5.215*
REG	−1.644	−4.434*	−2.240	−4.414*
TRA	−1.611	−4.872*	−2.236	−4.914*
Westerlund test for cointegration among panels				
Statistic	value	Z-statists	p-value	Robust p-values
Gt	−4.11	−3.798	.0.001	0.005
Ga	−14.867	−3.421	0.000	0.002
Pt	−9.532	−4.021	0.003	0.006
Pa	−22.104	−3.988	0.000	0.001

Note. *, ** and *** denote level of significance at 1%, 5% and 10% confidence levels.

Table 6
CS-ARDL estimation (CO2 Emission-Dependent variable).

Variable	Coefficient	Std. Error	Z	P>|Z|
ECT (−1)	−0.441	0.068	−6.37	0.000
l.lCO2	0.356	0.048	7.29	0.000
L2.lCO2	−0.127	0.041	−3.15	0.000
d.lnGDPP	0.078	0.018	3.88	0.001
d. $\ln GDPP^2$	−0.020	0.007	2.56	0.011
d.FDI	0.009	0.006	1.62	0.017
d.TRA	0.016	0.005	2.38	0.008
d.INST	−0.011	0.019	−1.43	0.016
d.meanCO2	0.048	0.031	2.68	0.007
d.meanGDPP	0.023	0.014	2.25	0.024
d.meanFDI	−0.004	0.002	−1.79	0.051
d.meanTRA	0.008	0.004	2.10	0.046
d.meanREG	−0.011	0.015	−2.26	0.031
d.meanINST	0.027	0.002	1.51	0.013
Long-run coefficients				
lnGDPP	0.318	0.082	3.88	0.000
$lnGDPP^2$	−0.005	0.006	−1.64	0.051
FDI	−0.015	0.008	−1.94	0.055
TRA	0.071	0.025	2.86	0.004
INST	−0.089	0.045	−1.96	0.050
Hausman Test Results PMG vs MG (CS-ARDL framework)				
Chi-square statistic		3.83		
Degrees of freedom		6		
P-value		0.443		

developing country often fuels economic expansion, which tends to increase fossil fuel energy consumption. This proposition is supported by Nguyen and Duong (2025), who posit that an increase in pollution among developing countries is often due to economic expansion that follows increased FDI inflow, especially in countries with poor institutional quality. Reinforcing these findings, Liu and Xu (2024) revealed that FDI inflow into developing countries, especially through China's Belt and Road Initiative (BRI), often leads to a shift in multinational enterprises into countries with low regulations on environmental conservation. Furthermore, Aydin and Destek (2025) posit that FDI inflow, along with financial development, always increases CO2 emission unless it is accompanied by stronger institutional quality frameworks and clean energy regulations.

Institutional quality, on the other hand, has a significantly negative effect in the long run (coefficient −0.089), which implies that better institutional quality reduces emissions among developing countries. This is consistent with the findings of Huang et al. (2024) and Gerged et al. (2023), who proposed that effective institutions enforce environmental policies that curb emissions, thereby lowering pollution. Furthermore, the negative effect of institutional quality on CO2 emissions may be due to better-designed, implemented, and enforced environmental regulations resulting from stronger institutions in the country. For instance, stronger institutional quality, characterized by regulatory quality, government effectiveness, rule of law enforcement, and corruption control, often creates an environment where policies such as industrial emission regulation and clean energy technology are enforced, leading to better performance (Chen, 2024).

This study also conducted a Hausman test to choose between the pooled mean group estimation and the mean group estimation, and presented the results in Table 6. The null hypothesis of this test is that the long-run coefficients are equal across panels; thus, PMG is preferred over an alternative hypothesis of heterogeneous slopes across panels. As such, the result provided for the test Chi-square statistic of 3.83 and a p-value of 0.443, implying that the null hypothesis is not rejected and hence PMG estimators are preferred to MG (Hamrouni et al., 2025). The test results support the homogeneity assumption across the selected sample in the long run while allowing heterogeneity in the short run. The economic implication of these findings is that variables such as economic growth, institutional quality and renewable energy consumption tend to exert similar pressure on CO2 emission in the long run across the selected developing countries (Olamide et al., 2022).

4.6. Mediating effect of renewable energy consumption on the growth-emission nexus

To determine whether renewable energy consumption in developing countries mediates the growth-emission path, three regression models were estimated, and the results are presented in Table 7. The models followed the CS-ARDL estimation technique with varying dependent and control variables. The first column of this table replicates the results of the CS-ARDL estimation model in Table 6, where CO2 emissions are the dependent variable and show significant positive and negative relationships with GDP per capita and squared GDP per capita. The second column presents a similar estimate, with renewable energy consumption as the dependent variable, as in Equation (3). This estimation found a positive and significant relationship between GDP per capita and its squared variable, and renewable energy consumption (Coefficients = 0.055 and 0.038, significant at 5%). In the third column, renewable energy consumption and GDP per capita were incorporated into the CS-ARDL model as independent variables, while CO2 emissions were the dependent variable. The result reveals a positive, significant relationship between renewable energy consumption (coefficient = 0.062) and GDP per capita (coefficient = 0.052) and emissions, while GDP per capita squared had a negative, significant effect on emissions. According to the proposition by Ojeka et al. (2024) and following this study's context, renewable energy consumption then mediates the nexus between growth and environmental quality if GDP growth per capita significantly influences renewable energy consumption, while environmental quality is also influenced by renewable energy consumption, as demonstrated in the models.

Since the coefficient of GDP per capita in the first column (0.078) is greater than its coefficient in the third column (0.052), and that of squared GDP per capita follows the same pattern, it can be concluded that renewable energy consumption mediates the relationship between growth and emissions. Specifically, the direct effect of growth on emissions is 0.052, while the mediating effect is 0.0032 (0.052×0.062), which is relatively low. Similarly, at higher income levels, the direct effect on growth is reported as −0.041, while the relative mediating effect is −0.0025 (0.01×0.062). The meaning is that GDP per capita growth not only directly influences CO2 emissions in the region but also promotes renewable energy consumption, which enhances environmental quality.

4.7. Threshold effect of economic growth on emissions: the EKC hypothesis

This study employed the dynamic panel threshold (kink) model (DPTM) proposed by Seo and Shin (2016) to determine the minimum level of GDP per capita required to enhance environmental quality, as proposed by the EKC hypothesis. As such, Equation (7) was estimated, and the results are presented in Table 8. From this table, we find that lagged CO2 emissions are statistically significant at the 1% level, which justifies the selection of a dynamic threshold model, rather than static threshold models such as Hansen (1999), for estimating the relationship between the study variables. The incorporation of lagged dependent variables in the threshold estimation as control variables also helps address endogeneity concerns in the model, allowing the estimation of unbiased, consistent estimators (Sikhawal, 2023). Similarly, the kink model estimation was a true dynamic panel threshold model for the selected panels, as implied by a statistically significant kink slope coefficient at the 5% level. The negative coefficient of the kink slope (coefficient = −1.843) implies that, beyond the estimated threshold level of income, increases in GDP per capita result in reduced emissions, which is consistent with the EKC hypothesis. Furthermore, the Wald test for continuity of restrictions yielded an insignificant *Chi*2 statistic at

Table 7
Mediating effect test.

Variables	(1)	(2)	(3)
	lnCO2	RE	lnCO2
lnGDPP	0.078***	0.055**	0.052**
lnGDPP^2	−0.020**	0.038**	−0.041**
RE			0.062**
Control variables	Yes	Yes	Yes

*** and ** denote significance level at 1 and 5% respectively.

Table 8
The threshold effect of economic growth on environmental degradation.

Dep var: lCO2	Coefficient	Std. Error	Z	P>|Z|
l.lnCO2	0.432	0.012	31.41	0.000
lnGDP (> γ)	−0.034	0.02	−3.64	0.000
FDI	0.011	0.008	2.87	0.007
TRA	0.006	0.002	2.92	0.004
REG	−0.018	0.004	−3.44	0.001
INST	−0.009	0.001	−2.43	0.015
Kink-slope	−1.843	0.752	−2.45	0.014
Threshold value (γ)	8.35	0.008	40.46	0.000
Test for continuity restriction (Kink vs Step model)				**Prob > Chi2**
Bootstrap replications	500			
Prob > Boots	0.000			
Wald test	2.87			0.091
LR-test	2.95			0.083

the 5% level, suggesting we cannot reject the null hypothesis of continuity. This suggests that the kink specification model is preferred over the unrestricted model, which is consistent with the assumption of this study that GDP per capita exerts a continuous, though slope-changing, effect on emissions around the estimated threshold level (Seo et al., 2019).

The threshold variable (GDP per capita) is statistically significant and negative above the threshold level (coefficient = −0.034) at the 1% level, which confirms a threshold relationship between growth and emissions. The threshold value is estimated at ≈$4230.18 per annum, obtained by taking the antilogarithm (exponential) of $\gamma = 8.35$. This threshold level is statistically significant, confirming the threshold relationship between growth and CO2 emissions (Giannerini et al., 2024). The economic implications of this result are that higher per capita growth promotes environmental sustainability. This could be because, in the early stages of industrialization, economic diversification and a shift from agriculture to manufacturing and service industries often lead to increased pollution and environmental degradation, which is corrected as national income rises. A notable example is Mauritius: between 1980 and 2020, GDP per capita increased from USD 2100 to USD 9,700, representing over 300% growth in four decades. This saw the country transform from an initial monocrop economy to a diversified economic system driven by manufacturing, tourism, and financial services (Tandrayen Ragoobur & Narsoo, 2022). This economic shift prompted the country to invest in environmental protection projects and policies, such as an integrated solid waste management system and renewable energy consumption targets, under the Environmental Protection Act. These policies led to improvements in water and air quality, reduced deforestation, and the conservation of the forest ecosystem, thereby reducing pollution. In line with this finding, developing countries should intensify investment in human capital through education and skills training, as well as providing innovation incentives to boost income growth.

4.8. Moderating effect of institutional quality consumption on growth-emission nexus

Institutional quality was introduced into the model as a moderating variable to examine whether it accelerates the attainment of the EKC threshold point. As such, this study estimated Equation (8), where the threshold variable is the interaction between institutional quality and GDP growth per capita (lnGDP*INT), and presented the results in Table 9. The result from this table indicates that the interaction term has a statistically negative effect on emissions at 1%, implying that improvements in institutional quality dampen the emission-intensive effect of economic growth (Hamrouni et al., 2025). Similarly, a negative and significant kink slope coefficient (coefficient = −2.017) further implies that, beyond the estimated threshold level, the interaction between GDP per capita and institutional quality significantly reduces emissions. Furthermore, the Wald test for this result provides an insignificant Chi^2 statistic at 5% level of significance, which supports the suitability of the kink continuity model and the slope-changing effect around the estimated

Table 9
To test whether institutional quality accelerates the threshold relationship.

Dep var: lCO2	Coefficient	Std. Error	Z	P>|Z|
l.lnCO2	0.724	0.024	29.58	0.000
lnGDP*INT (> ρ)	−0.051	0.015	−4.08	0.000
FDI	0.012	0.007	2.48	0.013
TRA	0.052	0.016	3.21	0.001
REG	−0.017	0.049	−3.53	0.000
Kink-slope	−2.017	0.801	−2.52	0.012
Threshold value (r)	7.81	0.009	29.41	0.000
Test for continuity restriction (Kink vs Step model)				**Prob > Chi2**
Bootstrap replications	500			
Prob > Boots	0.000			
Wald test	2.22			0.101
LR-test	2.25			0.093

threshold value.

The threshold level of interaction between GDP per capita and institutions was estimated at ≈ $2465.13, taking the anti-logarithm of $\rho = 7.81$, which is below the threshold value of GDP per capita as a standalone variable at $\approx 4,230.18$ UDS. This means that integration of institutional quality with economic growth accelerates the attainment of the Environmental Kuznets Curve turning point**.** This is because stronger institutional quality, when combined with higher economic growth, promotes proper governance, innovation, and policy enforcement, which accelerates the EKC hypothesis by surpassing the turning point much earlier (Azimi & Rahman, 2024). As pointed out by Nguyen and Duong (2025), when economic growth occurs in a high institutional quality environment, it is likely to be environmentally sustainable, guided by increased public awareness, well-enforced environmental policies and standards, and proper allocation of resources toward investment into cleaner energy, which accelerates the EKC turning point. Empirically, the findings of this study are supported by previous studies such as Nguyen et al. (2023), who also found that stronger institutional quality moderates the relationship between income growth and environmental pollution through enhancing environmentally friendly policies, thus accelerating the turning point of the Environmental Kuznets Curve at a lower income level. On the other hand, weak institutions tend to prolong environmental degradation despite greater economic growth because of a lack of policy enforcement mechanisms (Dina et al., 2021).

The validity of these threshold estimation results was further tested using Seo and Shin's (2016) bootstrap test for both linearity and threshold effect. The null hypothesis of the linearity test provides that the estimated model follows a linear pattern, while that of the threshold test assumes no threshold effect in the model. From Tables 8 and 9, the null hypothesis of linearity and no threshold effect is rejected at the 1% level, confirming the validity of the estimation model. This result implies that the relationship between GDP per capita and emissions is non-linear and has a threshold point beyond which an increase in GDP changes the sign and effect on emissions among developing countries. Similarly, consistency in results obtained from Tables 6 and 7, as well as 8 and 9, in terms of significance levels and coefficient signs, suggests that this study's results are robust.

5. Conclusion and policy recommendations

This study deviates from the conventional growth-emission literature to analyze the role of renewable energy consumption and institutional quality in the EKC test as mediating and moderating variables, respectively. Using a sample of 77 developing countries for the period between 200 and 2023, this study employed CS-ARDL and dynamic panel threshold (kink) models to estimate the non-linear relationship between growth and emission, considering the role played by renewable energy consumption and institutional quality in this path. The novelty of the current study thus lies in determining the mediating role of renewable energy consumption in the growth–emission nexus and checking whether the institutional quality accelerates the attainment of the EKC turning point. The estimated results were found consistent in terms of variable coefficients as well as significance across models, implying they are robust.

The primary results from the CS-ARLD model reveal that, in the early stages of growth, economic growth increases emissions in developing countries, which is corrected when high growth is achieved, and that renewable energy consumption determines the path from growth to emissions. The threshold analysis further supports CS-ARD results by revealing the existence of a threshold level GDP per capita beyond which further increase in economic growth reduces emissions, estimated at $\$4,230.18$. Beyond this threshold level, the coefficient of GDP per capita is negative, implying an improvement in environmental quality, thus a threshold effect. The analysis of the moderating role of institutional quality further reveals that the threshold level of economic growth required for environmental improvement is reached earlier when proper institutions are in place, estimated at $2465.13. This finding underscores the role of stronger institutional qualities in promoting environmental conservation among developing countries. This study, therefore, concludes that renewable energy consumption mediates the growth and emission nexus while institutional quality moderates the relationship by accelerating the EKC turning point.

Based on the above conclusion, the following policy recommendations are proposed;

1) Developing countries to scale up investment in renewable energy infrastructure to accelerate its adoption through fiscal incentives, competitive auctions, and streamlined regulatory approvals that will attract private investment. In particular, governments can opt for redirecting inefficient fossil fuel subsidy towards financing renewable energy investment in order to strengthen energy security and lower environmental cost attributed to economic growth.
2) Strengthening institutional quality for the enforcement of environmental policies. Policy makers in developing countries should enhance institutional effectiveness through enforcing environmental regulations, improving governance transparency, and reducing corruption. For instance, the establishment of independent regulatory agencies, digital monitoring systems, and an accountable public procurement system can be tools to enhance compliance in the pollution compliance sector. Strong institutions will attract green investment by reducing policy uncertainty, thus ensuring economic growth translates into improvement in environmental quality rather than stimulating pollution.
3) Promote income per capita growth through structural transformation. Since the reduction in emissions starts to be experienced only after income reaches a certain level, policymakers should focus on promoting investment in human capital to foster income growth and a sustainable environment. For instance, actions such as supporting green manufacturing, expanding technical training, and promoting climate-smart agriculture can be used to raise productivity and income while limiting emission intensity. Aligning industrial policies with sustainable environmental policies will enable developing countries to surpass the threshold level of income per capita while also reducing environmental degradation.

In addition to the stated policy recommendations, this study extends the existing literature on the economic growth and environmental quality nexus by uniquely establishing a framework that integrates the role of mediating and moderating variables in the context of developing countries. Methodologically, this study is among the first studies to use the robust cross-sectional augmented autoregressive distributive lag (CR-ARDL) model alongside the dynamic panel threshold kink model to determine the Environmental Kuznets Curve's turning point for a large panel data set for 77 developing countries spanning 200-2023. The moderating effect of institutional quality on the EKC turning point provides the clearest empirical evidence on the need for differential policy efficacy for environmental sustainability. This shifts the EKC debate from a mere existence to the determination of how to accelerate it.

Despite the contributions, the current study also has some limitations that future studies can address. First, the current study exclusively uses CO2 emissions as a proxy for environmental degradation. While the variable properly explains climate change, it is only one among other dimensions of environmental impact. Other pollution aspects such as deforestation, water pollution, loss in biodiversity and ecological footprints have not been considered by the current study. Second, the reliance on aggregate data by this study may obscure regional variation in terms of environmental policies and level of institutional quality. Therefore, future studies can estimate this relationship considering other dimensions or composite index of environmental sustainability while also desegregating developing countries data into respective region to capture the regional differences.

Conflict of interest

There is no conflict of interest regarding the publication of this manuscript.

Data availability

Data will be made available on request.

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